BERKELEY TALK: COHERENT COHOMOLOGY OF SHIMURA VARIETIES, MOTIVIC COHOMOLOGY AND PERIOD INTEGRALS

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The main theme of conjecture of Venkatesh is that the same Hecke eigensystem appears in multiple degrees of cohomology of locally symmetric spaces. Our main goal is to formulate an analogous conjecture for Shimura varieties, when there are such multiple occurrences. Ultimately the conjecture will be of form

$$\wedge^* \mathcal{M} \otimes \mathcal{H}^{\min} \cong \mathcal{H}^*,$$

where $\mathcal{M}$ is some **"adjoint motivic cohomology"** and $\mathcal{H}^*$ is **"coherent cohomology of Shimura varieties**. It turns out that apperance in multiple degrees only happens when the weight is singular, and this involves several interesting features.

1. Modular forms

We start by sketching the most basic case, the case of modular forms **(of weight one)**, in a reasonable detail. Recall that weight one modular forms are sections of line bundle ω over a modular curve X, so that the space of weight one modular forms is identified with

$$M_1 = H^0(X, \omega).$$

Explicitly, they are identified as $f \leftrightarrow f\alpha$, where $\alpha = 2\pi i dw$ (corresponds to dt/t over $\mathbb{C}^\times/q^{\mathbb{Z}}$, $q = \exp(2\pi i w)$; this shows this normalization respects the q-expansions) is a canonical relative differential of the universal elliptic curve $E \to \mathbb{H}$, where over $\tau \in \mathbb{H}$, $E_\tau = \mathbb{C}_w/(\mathbb{Z} + \mathbb{Z}\tau)$.

This admits the action of Hecke operators. As the modular curve is defined over $\mathbb{Q}$, M_1 has a natural $\mathbb{Q}$-structure, and the Hecke action is also defined over $\mathbb{Q}$. In familiar terms, the $\mathbb{Q}$-rational structure is characterized by the space of modular forms with Fourier coefficients (maybe except the constant term) in $\mathbb{Q}$. Consider the space of cusp forms $S_1 \subset M_1$, which also enjoys the same properties.

On the other hand, it is thought that $H^1(X, \omega)$ also represents the space of weight 1 cusp forms, via

$$H^1(X, \omega) \underset{\text{Serre duality}}{\cong} H^0(X, \omega(-D))^\vee = S_1^\vee \underset{\text{Petersson inner product}}{\cong} S_1.$$

More explicitly we can give an isomorphism as follows. Note that by Hodge theory $H^1(X, \omega)$ can be computed using $(0,1)$-forms on X, namely by the Dolbeault complex

$$0 \to H^0(X, \omega) \to \mathscr{A}^{0,0}(X, \omega) \to \mathscr{A}^{0,1}(X, \omega) \to H^1(X, \omega) \to 0,$$

where $\mathscr{A}^{0,i}(X, \omega)$ can be regarded as the space of Γ-invariant (Γ is the level) C^∞-sections of $\omega \otimes \overline{\Omega^i}$ on $\mathbb{H}$, the upper half plane. We have some obvious choices for sections of each line bundles: α for ω, and $d\overline{z}$ for $\overline{\Omega^1}$. One readily checks[1] that

$$\widetilde{f} := \frac{\overline{f}(z)\alpha d\overline{z}}{y},$$

[1]Because of y^{-1}, one can only do this for cuspidal f.

is a Γ-invariant section of $\omega \otimes \overline{\Omega^1}$.

Indeed, if we say provisionally that a smooth function h on $\mathbb{H}$ is of weight (m,n) if $h(\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) z) = (cz+d)^m(c\overline{z}+d)^n h(z)$ for $\gamma \in \Gamma$, then α is weight $(-1,0)$, $d\overline{z}$ is weight $(0,-2)$, and y^{-1} is weight $(1,1)$. Also, since f is weight $(1,0)$, $\overline{f}$ is weight $(0,1)$. Now one checks

$$(-1,0)+(0,-2)+(1,1)+(0,1)=(0,0).$$

By the same reason as $H^0(X,\omega)$, $H^1(X,\omega)$ has a $\mathbb{Q}$-rational structure coming from the geometry of X and ω.

Q. How are the two rational structures related?

One obvious way to relate them is to use **"the" Serre duality**,

$$H^0(X_k,\omega(-D)) \otimes H^1(X_k,\omega) \to H^1(X_k,\Omega^1) \underset{\text{trace}}{\xrightarrow{\sim}} H^0(X_k,\mathcal{O}) = k,$$

where X_k is the modular curve seen as an algebraic curve over a field k. In particular, this works for $k=\mathbb{Q}$. It turns out that "the" Serre duality is in this case

$$p(z)dz \wedge d\overline{z} \mapsto \frac{1}{2\pi i}\int_X p(z)dz \wedge d\overline{z},$$

where $p(z)dz \wedge d\overline{z}$ is a Dolbeault class in $\mathscr{A}^{0,1}(X,\Omega^1)$.

"The" Serre duality is compatible with the $\mathbb{Q}$-rational structures of coherent cohomology, so it can be used to compare the $\mathbb{Q}$-rational structures of H^0 and H^1. Namely, if $f \in H^0(X,\omega)$ is a holomorphic cuspform of weight one with $\mathbb{Q}$-rational Fourier coefficients, then there is $c \in \mathbb{C}^\times$ such that $c\widetilde{f} \in H^1(X,\omega)$ is defined over $\mathbb{Q}$. Then, by "the" Serre duality,

$$\langle f, c\widetilde{f}\rangle_{Serre} \in \mathbb{Q}.$$

On the other hand, by the formula of "the" Serre duality, we have

$$\langle f, c\widetilde{f}\rangle_{Serre} = \frac{1}{2\pi i}\langle f, \overline{c}f\rangle_{Petersson} = \frac{c}{2\pi i}\langle f,f\rangle_{Petersson}.$$

Thus,

$$c \in \frac{2\pi i}{\langle f,f\rangle_{Petersson}}\mathbb{Q}.$$

On the other hand, by the Rankin–Selberg method,

$$L(s, f\otimes g) = \int_{\Gamma\backslash\mathbb{H}} f(z)\overline{g(z)E(s,z)}y^{-1}dxdy,$$

for some Eisenstein series $E(s,z)$, which has a simple pole at $s=1$ with residue $\in \mathbb{Q}$. Thus,

$$\operatorname{Res}_{s=1} L(s, f\otimes f) \sim_{\mathbb{Q}^\times} \langle f,f\rangle_{Petersson},$$

where $\sim_{k^\times}$ means that both sides are equal up to a factor of $k^\times$. Since

$$L(1,f,\mathrm{Ad}) \sim_{\mathbb{Q}^\times} \pi^2 \operatorname{Res}_{s=1} L(s, f\otimes f),$$

we have

$$L(1,f,\mathrm{Ad}) \sim_{\mathbb{Q}^\times} \pi^2\langle f,f\rangle_{Petersson}.$$

Everything up to this point is not special to weight one, and applies to modular forms of general weight. However, there is a special feature to weight one modular forms, that the Galois representation attached to f is of finite image **(Artin representation)**, so there is a special conjecture relating $L(1, f, \mathrm{Ad})$ with the arithmetic of f, called the **Stark conjecture**.

2. Stark conjectures

The Stark conjecture is about special value of L-function attached to an Artin representation, or a continuous finite-dimensional representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ with complex coefficients (which is necessarily of finite image, as complex numbers have no infinite profinite subgroup).

Conjecture 2.1 (Stark conjecture). [2]*Let $\rho : G_{\mathbb{Q}} \to \mathrm{GL}_n(\mathbb{C})$ be a continuous representation which does not have the trivial representation as its constituent. Then, there is an explicit integer*[3] *m such that*

$$L(1, \rho) \sim_{\overline{\mathbb{Q}}^\times} \pi^{n-m} R(\rho),$$

*where $R(\rho)$ is the **Stark regulator** defined as follows.*

- *As ρ is of finite image, ρ factors through $\mathrm{Gal}(L/\mathbb{Q})$ for some $[L : \mathbb{Q}] < \infty$. Pick an embedding $L \hookrightarrow \mathbb{C}$ which gives rise to a complex conjugation $c \in \mathrm{Gal}(L/\mathbb{Q})$.*
- *The group of units $U_L^\times$ have rank $\#|\mathrm{Gal}(L/\mathbb{Q})/\langle c\rangle| - 1$ by Dirichlet unit theorem. This is in fact **"generated by one unit"**, in a sense that there is a unit ϵ fixed by c such that U_L is (up to torsion) spanned by $\sigma(\epsilon)$ for $\sigma \in \mathrm{Gal}(L/\mathbb{Q})/\langle c\rangle$, and there is one relation that*

  $$\prod_{\sigma \in \mathrm{Gal}(L/\mathbb{Q})/\langle c\rangle} \sigma(\epsilon) = \pm 1.$$

  In particular, $U_L/U_L^{\mathrm{tor}} = [\mathrm{Ind}_{\langle c\rangle}^{\mathrm{Gal}(L/\mathbb{Q})} \mathbb{Z}] - [\mathbb{Z}]$ in the K-group of $\mathbb{Z}$-representations of $\mathrm{Gal}(L/\mathbb{Q})$.
- *Now the **Stark units** are defined as*

  $$U_L[\rho] = (\rho_{\overline{\mathbb{Q}}} \otimes_{\mathbb{Z}} U_L)^{\overline{\mathbb{Q}}[\mathrm{Gal}(L/\mathbb{Q})]},$$

  where $\rho_{\overline{\mathbb{Q}}} : G_{\mathbb{Q}} \to \mathrm{GL}_n(\overline{\mathbb{Q}})$ is such that $(\rho_{\overline{\mathbb{Q}}}) \otimes_{\overline{\mathbb{Q}}} \mathbb{C} = \rho$[4]*. Thus by above point and Frobenius reciprocity,*

  $$U_L[\rho] \cong \rho_{\overline{\mathbb{Q}}}^{\langle c\rangle}.$$
- *The **Stark regulator** is the (co)volume of the $\overline{\mathbb{Q}}$-lattice $\rho_{\overline{\mathbb{Q}}}^{\langle c\rangle}$ inside $\rho^{\langle c\rangle}$, where the $\mathbb{C}$-basis of ρ is decided by the condition that*

  $$\rho(c) = \begin{pmatrix} I_m & \\ & -I_{n-m} \end{pmatrix}.$$

In general the Stark regulator is given as **"log of units"**.

[2]This is a weakening of actual Stark's conjecture, e.g. we do not care too much about field of definition being a number field.

[3]m is such that $\rho(c) \sim \begin{pmatrix} I_m & \\ & -I_{n-m} \end{pmatrix}$.

[4]ρ is of finite image so it can be descended to $\overline{\mathbb{Q}}$.

Example 2.2. An instance is given by the Dirichlet regulator

$$\mathcal{O}_F^\times \xrightarrow{(\log|\sigma(\epsilon)|)_\sigma} \mathbb{R}^{r_1+r_2},$$

which is when ρ is given by $\mathrm{Gal}(L/\mathbb{Q}) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$.

Now let $\rho = \mathrm{Ad}\,\rho_f$, where ρ_f is the Artin representation associated to the weight one modular form f. Then, $L(1, f, \mathrm{Ad}) \sim_{\overline{\mathbb{Q}}^\times} \pi^2 R(\rho)$, or

$$R(\rho) \sim_{\overline{\mathbb{Q}}^\times} \langle f, f\rangle_{Petersson},$$

or

$$c \sim_{\overline{\mathbb{Q}}^\times} \frac{\pi}{R(\rho)}.$$

It is now connected to **motivic cohomology** as the group of Stark units is (conjecturally) identified with the motivic cohomology group of adjoint motive,

$$H^1_{\mathcal{M}}((\mathrm{Ad}\,\rho)_{\mathbb{Z}}, \overline{\mathbb{Q}}(1)) \cong U_L[\rho].$$

[5]The Stark regulator $\rho^{\langle c\rangle}_{\overline{\mathbb{Q}}} \to \rho^{\langle c\rangle}$ is in this case the **Beilinson regulator** of adjoint motive,

$$H^1_{\mathcal{M}}((\mathrm{Ad}\,\rho)_{\mathbb{Z}}, \overline{\mathbb{Q}}(1)) \to H^1_{\mathscr{D}}((\mathrm{Ad}\,\rho)_{\mathbb{R}}, \mathbb{C}(1)).$$

Thus, $R(\rho)$ is a **special case of the volume of motivic cohomology via Beilinson regulator**.

3. Formulation of the main conjecture

We have massaged the formulation of the weight one modular form case enough so that we can generalize to other Shimura varieties. We fix a reductive quasisplit group $G/\mathbb{Q}$ that gives rise to a Shimura variety, and a cuspidal tempered automorphic representation $\Pi = \Pi_f \otimes \Pi_\infty$ where Π_∞ is of **singular weight**; this means the infinitesimal character lies on a wall of Weyl chambers.

In terms of coherent cohomology of Shimura varieties, assume that an automorphic form in Π "appears" in H^i of Shimura variety, with coefficient bundle V.we consider

$$H^i(X)[\Pi_f] := \varinjlim_{K\subset G(\mathbb{A}_f)} H^i(X_G(K), V)[\Pi_f],$$

where the direct limit admits an action of $G(\mathbb{A}_f)$ and $[\Pi_f]$ is the Π_f-isotypic summand.

Similar to the case of weight one modular forms, as Π_∞ is of singular weight, $H^*(X)[\Pi_f]$ is nonzero in multiple degrees.

Conjecture 3.1. *As $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian metric,*

$$H^*(X_{\overline{\mathbb{Q}}})[\Pi_f] \cong H^{\max}(X_{\overline{\mathbb{Q}}})[\Pi_f] \otimes \wedge^*\mathcal{M},$$

where

- *$X_{\overline{\mathbb{Q}}}$ is the canonical model of X over $\overline{\mathbb{Q}}$, and $H^*(X)[\Pi_f]$ admits a Hermitian metric by identifying with harmonic Dolbeault forms (and the Dolbeault forms have Hermitian inner product coming from Hermitian metric on X and V),*
- *$H^{\max}$ means H^i where $i = \max_i\{H^i \neq 0\}$,*
- *$\mathcal{M} = H^1_{\mathcal{M}}((\mathrm{Ad}\,M)_{\mathbb{Z}}, \overline{\mathbb{Q}}(1))$, where $\mathrm{Ad}\,M$ is the **"adjoint motive"** where M is the "motive associated to Π" via Langlands philosophy*

[5]This implicitly uses that $U_L[\rho]$ does not depend on L which is easy to check.

- *$\mathcal{M}$ is equipped with Hermitian metric after giving a group-theoretic Hermitian metric (e.g. Killing form) on its archimedean realization (absolute Hodge cohomology) via Beilinson regulator, as*
$$H^1_{\mathscr{D}}((\operatorname{Ad} M)_{\mathbb{R}}, \mathbb{C}(1)) = \widehat{\mathfrak{g}}^{W_{\mathbb{R}}},$$
where $W_{\mathbb{R}} \to \widehat{\mathfrak{g}}$ is the archimedean Langlands parameter.

Example 3.2 (Modular forms of weight one)**.** Indeed, each aspect of modular form of weight one case matches up in this picture.

This has several interesting remarks we mention in passing.

(1) (Archimedean L-packet) In reality, each H^i corresponds to **different infinity types**. For example, even in the case of weight one modular forms, H^0 corresponds to f, a weight one holomorphic modular form, but H^1 corresopnds to $\overline{f}$, a weight negative one **anti-holomorphic modular form**. Thus, this conjecture is about comparing periods of **different automorphic representations** with changing infinity types in an archimedean L-packet.

(2) (Generalized complex conjugation, p-adic stuff) As we are comparing different automorphic representations, we need to be careful about **how to simultaneously normalize different automorphic representations**. For example, we want the conjecture to say that, for weight one modular forms, if f is algebraic (i.e. defined over $\overline{\mathbb{Q}}$), then $\frac{\pi \overline{f}}{\operatorname{vol} H^1_{\mathcal{M}}(=R(\rho))}$ is "algebraic" (i.e. corresponding coherent cohomology class in H^1 is defined over $\overline{\mathbb{Q}}$), but there is really no justification to use $\overline{f}$ over any other scalar multiple of $\overline{f}$ (this lies in a $\mathbb{C}$-line but we need a vector). We use metric structure to resolve some ambiguity but we still do not know how to exactly pin down complex conjugate.
   (a) In a more general case, the analogue of complex conjugation, dubbed **generalized complex conjugations**, should send an automorphic form of one infinity type to an automorphic form of another infinity type in the same archimedean L-packet. We hope that we can have a say on what this is in the future using Penrose transform.
   (b) This is also important in a view towards other realizations of motivic conjectures, say the conjecture of Harris–Venkatesh: how do you see the complex conjugate of a newform p-adically?

The Beilinson's conjectures give a conjectural answer to the volume of $\mathcal{M}$ in terms of L-values and some other invariant related to Betti and de Rham realization of $\operatorname{Ad} M$. Thus, we can formulate the subset of the conjecture on the top and bottom degrees

Conjecture 3.3. *If $f_i \in H^i(X_{\overline{\mathbb{Q}}})[\Pi_f]$, then*
$$\frac{\langle f_{\min}, f_{\min}\rangle_{Petersson}}{\langle f_{\max}, f_{\max}\rangle_{Petersson}} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} (\textit{Archimedean factor}) \left| \frac{L(1, \Pi, \operatorname{Ad})}{\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} \Pi)} \right|^2,$$
where the Archimedean factor is an explicit power of π, and $\langle f, f\rangle_{Petersson}$ is the Petersson inner product (integral of $|f|^2$ over the Shimura variety with invariant metric).

The above conjecture is equivalent to the main conjecture under the Beilinson's conjectures.

Example 3.4. Indeed, if Π is a weight one modular form, then $\operatorname{Ad} \Pi$ is pure of weight zero, so $F^1 H_{\mathrm{dR}}(\operatorname{Ad} \Pi) = 0$, and we are only left with $L(1, \Pi, \operatorname{Ad})$ term, which is what we've observed above.

Remark 3.5. The top and bottom degrees are not in general dual to each other. Rather, they depend on the location of the infinitesimal character in the Weyl chambers.

4. Cohomological period integrals

Recall that we compared rational structures of coherent cohomology of different degrees using Serre duality, which has

- **cohomological interpretation** (Serre duality),
- and also **automorphic interpretation** (Petersson inner product, which is related to $L(1, \mathrm{Ad})$).

Thus, we might have a handle of the conjecture if we have such tool in other settings. It turns out that there are several examples that are **not Serre duality**. They all follow the pattern of

$$\int_{[H]} f(h)E(h)dh,$$

where $H \subset G$ is a sub-reductive group, f is a cusp form on G, and E is an Eisenstein series on H. On one hand, if f "appears" in $H^{\dim X_H}(X_G)$, then one can interpret this period integral as a cup product of coherent cohomology,

$$H^{\dim X_H}(X_G) \times H^0(X_H) \xrightarrow{(\mathrm{id}, i_*)} H^{\dim X_H}(X_G) \times H^{\dim X_G - \dim X_H}(X_G) \xrightarrow{\text{Serre duality}} \mathbb{C},$$

where $i : X_H \to X_G$ is the embedding of Shimura varieties. On the other hand, this period integral is related to a certain L-function via some variant of unfolding process for Rankin–Selberg integrals, when f admits a certain (degenerate) Whittaker model.

- $\mathrm{SL}_2 \times \mathrm{SL}_2 \subset \mathrm{Sp}_4$ (Novodvorsky) gives rise to the degree 4 spinor L-function of generic cusp forms of Sp_4.
- $\mathrm{SU}(1,1) \times \mathrm{SU}(1) \subset \mathrm{SU}(2,1)$ (Gelbart–Piatetski-Shapiro) gives rise to the base change L-function of generic cusp forms of $\mathrm{SU}(2,1)$.

From the two interpretations, the cohomological period integrals can relate

- **cohomological datum** (namely, algebraicity of f as coherent cohomology class) with
- **automorphic datum** (namely, Whittaker coefficients of f).

Using these, we prove the following.

Theorem 4.1. *Assume standard conjectures on periods. Then, in the above cases where $G = \mathrm{Sp}_{4,\mathbb{Q}}$ or $\mathrm{SU}(2,1)$, when $H^*(X)[\Pi_f]$ appears in H^0, H^1, then the conjecture is true.*

The standard conjectures are the following.

- Beilinson's conjectures, to relate volume of motivic cohomology with L-values.
- Gan–Gross–Prasad conjectures, to relate Petersson inner product of algebraic holomorphic forms (H^0) with L-values. This uses that Fourier coefficients of holomorphic forms detect algebraicity, and the Fourier coefficients are also described in terms of Bessel/Fourier–Jacobi period.
- Lapid–Mao conjectures, to relate Petersson inner product of Whittaker-algebraic generic forms (H^1) with L-values.